Algebra Practice Problems

Name: UID:

- Basics

  * $a - (-b) = a + b$
  * $a(b + c) = ab + ac$
  * $a + b = c$ can be rewritten as $a = c - b$ (Middle step : $a + b - b = c - b$)
  * $a \div b = a\left(\frac{1}{b}\right) = \frac{a}{b}$
  * $\frac{a/b}{c/d} = (a/b) \div (c/d) = (a/b)(d/c) = \left(\frac{a}{b}\right)\left(\frac{d}{c}\right) = \frac{ad}{bc}$

1. $.02 \times .3$
2. $\frac{.08}{.2}$
3. $\frac{2}{5} + \frac{3}{7}$
4. $\frac{2}{5} - (-\frac{3}{7})$
5. $\frac{5/7}{3/2}$
6. Solve $\frac{1}{3x-1} = \frac{1}{5}$.
7. Simplify $\frac{x}{2} - \frac{x}{3}$.
8. Expand $(x - 1)(x - 2)$.
9. Solve $\frac{1}{3x+2} = \frac{2}{4x-2}$.

- Exponents (Basics)

  * $x^{ab} = (x^a)^b$
  * $x^{a+b} = (x^a)(x^b)$
  * ax^b is **not** x^{ab} but $a \times x^b$. (e.g. $-x^{-2} = -\frac{1}{x^2}$ but never x^2.)
  * $x^{\frac{a}{b}} = (x^{\frac{1}{b}})^a$
  * $(-a)^b$
    when b is even : $(-a)^b = a^b$ (e.g. $(-2)^2 = 2^2 = 4$.)
    when b is odd : $(-a)^b = -a^b$
  * When solving $x^a = b$, get rid of the power a by using $(x^a)^{\frac{1}{a}} = b^{\frac{1}{a}}$. Then $x^{a\left(\frac{1}{a}\right)} = b^{\frac{1}{a}}$, and thus $x^1 = x = b^{\frac{1}{a}}$.
  * $x^a = b$ is the same thing as $\sqrt[a]{b} = x$.

1. Simplify $(x^2)(x^3)$.
2. Simplify $(x^2)^3$.
3. Solve $x^2 = 16$.
4. Solve $x^4 = 16$.
5. Solve $x^3 = 27$.
6. Solve $x^{\frac{3}{2}} = 4$.
7. Solve $\frac{2^{3x+4}}{2^{2x+3}} = 32$.

- Exponents (Full upgrade)
  - $x^{-r} = \frac{1}{x^r}$
  - $x^{\frac{1}{r}} = \sqrt[r]{x}$
  - $x^0 = 1$

1. Express the following expressions in the exponent forms (x^a for the right choice of a).
   i. $\frac{1}{x^2}$ ii. $\frac{1}{x}$ iii. $\sqrt{x}$ iv. $\sqrt[3]{x^4}$ v. $\frac{1}{\sqrt{x}}$
2. Simplify $\frac{x^2x^3}{x}$.
3. Simplify $\frac{a^2b^4c^{-2}}{a^2b^{-1}c}$.
4. Simplify $\frac{1}{3}3^{x+1}$.
5. Solve $\frac{x^{3/2}}{\sqrt{x}} = 5$.

- Factorization of quadratics
  1. $x^2 - x - 2$
  2. $x^2 + x - 2$
  3. $3x^2 + 3x - 6$
  4. $2x^2 + 5x + 3$